

Polynomials

A polynomial is a finite sum of terms whose variable exponents are nonnegative integers.

$$5x^3 - 2x + 8$$

What makes this expression a polynomial?

Testing the definition

$$x^{-1} + 4$$

Why is this expression not a polynomial in x?

Testing the definition

$$\sqrt{x} + 2$$

Why is this expression not a polynomial in x ?

Degree

The degree of a nonzero polynomial in one variable is its largest variable exponent.

$$3x^5 - 8x^2 + 6$$

What is the degree?

Leading coefficient

In descending-power order, the leading coefficient belongs to the highest-degree term.

$$9 - 2x^4 + 7x$$

What is the leading coefficient?

Standard form

$$6x - 4x^3 + 2x^2 - 9$$

How can we write this polynomial in descending-power order?

Counting terms

Monomials, binomials, and trinomials have one, two, and three nonzero terms, respectively.

$$4x^2 - 7x + 1$$

What name describes this polynomial by its number of terms?

Adding polynomials

$$(3x^2 - 2x + 5) + (x^2 + 6x - 8)$$

What is the sum in standard form?

Subtracting polynomials

$$(5x^2 + x - 4) - (2x^2 - 3x + 7)$$

What is the difference in standard form?

A negative sign before a sum

$$-(2x^2 - 5x + 3)$$

What equivalent polynomial has no parentheses?

A monomial times a polynomial

$$-3x(2x^2 - x + 4)$$

What is the product in standard form?

Two binomials

$$(x + 5)(x - 2)$$

What is the expanded product?

Two binomials with coefficients

$$(2x - 3)(3x + 4)$$

What is the expanded product?

A binomial times a trinomial

$$(x + 2)(x^2 - 3x + 4)$$

What is the product in standard form?

A squared binomial

$$(x - 4)^2$$

What is the expanded polynomial?

Conjugate binomials

$$(3x + 2)(3x - 2)$$

What is the expanded polynomial?

Several variables

The degree of a monomial is the sum of its variable exponents.

$$7x^2y^3$$

What is the degree of this monomial?

Combining multivariable terms

$$3x^2y + 5xy^2 - 2x^2y + xy^2$$

What equivalent expression combines the like terms?

A multivariable product

$$(x + 2y)(3x - y)$$

What is the expanded product?

An area expression

A rectangle has length $x + 4$ and width $x + 1$.

What polynomial in standard form represents its area?